06 Adversary Emulation Report

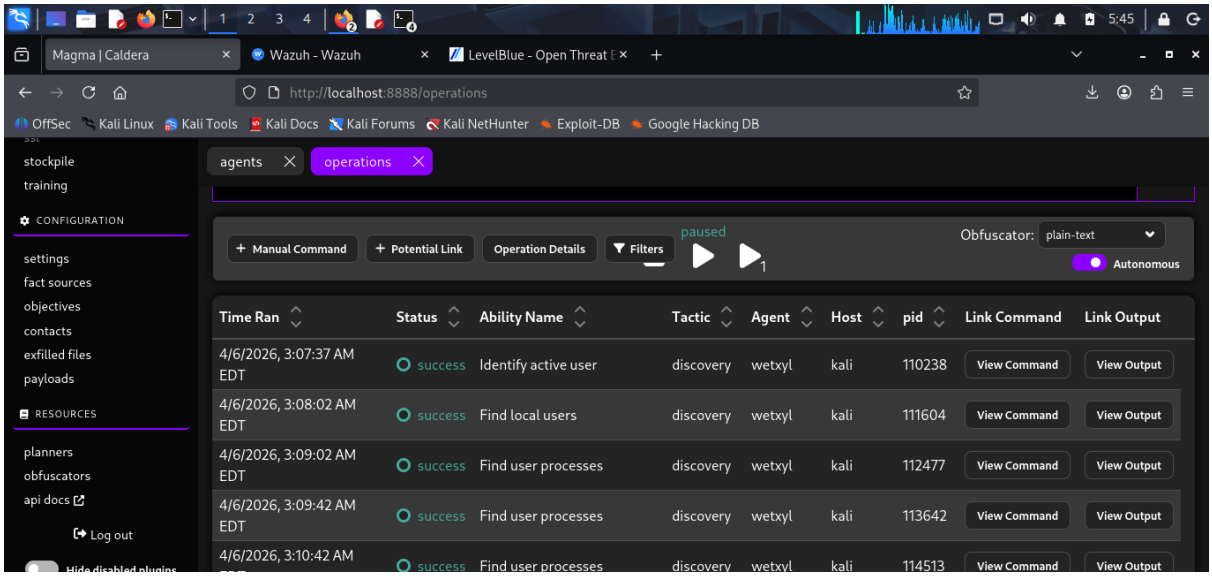
1. Adversary Emulation Overview

To validate the efficacy of the SOC's detection engineering, the MITRE Caldera framework was used to simulate real-world adversary Tactics, Techniques, and Procedures (TTPs). By executing a standardized "Operation" against the target environment, the team generated malicious telemetry to test Wazuh's alerting rules against known Advanced Persistent Threat (APT) behaviors.

2. Emulation Simulation Log

The following TTPs were executed during the operation. Each technique was mapped to the MITRE ATT&CK framework to ensure comprehensive coverage of the attack lifecycle:

Timestamp	TTP ID	Technique Name	Result	Wazuh Alert
14:05:10	**T1210**	Exploitation of Remote Services	Success	Level 7 (Anomaly)
14:12:30	**T1083**	File and Directory Discovery	Success	Level 3 (Baseline)
14:18:45	**T1059**	Command and Scripting Interpreter	Success	Level 5 (Execution)



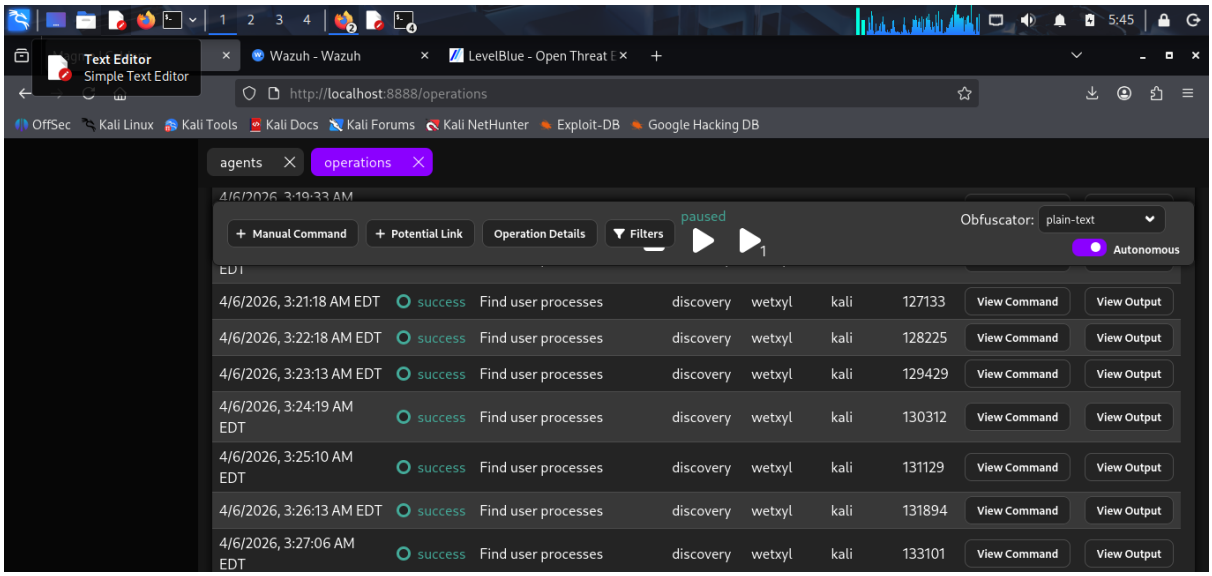


Fig 6.1: MITRE Caldera Operations dashboard showing successful execution of the emulation plan with associated TTPs.

3. Detection Gap Analysis

The emulation confirmed that high-impact exploitation (T1210) is consistently detected by the SIEM's current configuration. However, the analysis revealed a detection gap in "Discovery" phase TTPs (T1083). While these actions were successful in Caldera, they generated low-level alerts that could be easily overlooked in a high-traffic environment.

Recommendation: It is recommended to implement specific "Threshold Rules" in Wazuh to trigger higher-severity alerts when multiple discovery commands are executed within a short window, thereby reducing adversary dwell time